

Semester Project Ideas

Spring 2026

Guidelines for Semester Project:

- Projects are to be completed in teams of up to 3 members, using publicly available datasets, with focus on going beyond baseline implementations.
- Proper experimentation is expected, including comparisons with baselines and ablation studies to justify design choices.
- The mid-report and the final report must be submitted via LMS. Code must be uploaded on GitHub, and the repository link should be included on the **first page** of the final report.
- Some datasets are shared for reference; however, you are encouraged to explore and use a more suitable dataset of your own choice if needed.

Project Milestones and Timelines

Milestone	Deadline	Weight	Remarks
1. Team formation and project selection	19/04/26		Share details of your team members and your project via this form .
2. Mid report	08/05/26 (11:55 pm)	25%	Include a brief literature review, proposed methodology/architecture, and progress so far. Some preliminary results MUST be provided. Submit report via LMS.
3. Final report and code submission	22/05/26 (11:55 pm)	50%	Final report must follow ICML-style (4–5 pages) including: Abstract, Introduction, Literature Review, Methodology, Results, Ablations, and Conclusion. A statement on the usage of AI tools is mandatory. Submit report via LMS. Code must be uploaded on GitHub.
4. Presentation	25/05/26	25%	10-minute presentation summarizing methodology and results, followed by Q&A.

Project Ideas

01 Deepfake Audio Detection: Real vs. Synthetically Cloned Voices

Audio

Supervised

Binary Classification

With the rise of neural voice cloning systems, it has become increasingly easy to synthesize speech that sounds indistinguishable from a real human speaker. This project challenges students to build a classifier that determines whether a given audio clip is genuinely spoken by a human or generated by a text-to-speech or voice conversion model. The core difficulty lies not in achieving high training accuracy, but in generalization: models must perform well on spoofing attacks they have never seen before. Students will explore data augmentation on audio signals, analyze failure modes across different types of synthesis methods, and reason about the robustness and societal implications of anti-spoofing systems.

Dataset: **ASVspoof 2019** - Logical Access track. Contains over 121,000 utterances from 107 speakers, labeled as genuine or spoofed using 19 different TTS and voice conversion systems. Fully open download at <https://datashare.ed.ac.uk/handle/10283/3336>

02 Predicting Perceived Safety from Urban Street-View Imagery

Computer Vision

Supervised

Regression

Urban environments carry rich visual signals about neighborhood character, walkability, and perceived safety - cues that humans pick up on instantly but are difficult to quantify. This project tasks students with training a regression model to predict human-annotated safety scores from street-level images of cities and learn to handle a continuous regression target rather than class labels. Beyond the modeling, this project opens up important discussions on algorithmic fairness: the training labels reflect human perception, which is shaped by socioeconomic and racial biases, and students are expected to analyze and discuss the ethical implications of deploying such a model. Grad-CAM visualizations should be used to inspect what the model is attending to in the image.

Dataset: **MIT Place Pulse 2.0** - Contains 1.17 million pairwise comparisons of street-level images across six perceptual attributes including safety, wealth, and liveliness, crowdsourced from 81,000 volunteers.

03 Emotion Recognition from EEG Brain Signals

Time Series

Supervised

Multi-class Classification

Electroencephalography (EEG) records the brain’s electrical activity through scalp electrodes and encodes rich information about cognitive and emotional states. This project asks students to classify human emotional states - specifically valence (positive vs. negative) and arousal (excited vs. calm) - from raw multi-channel EEG recordings captured while subjects watched music videos. Students must design a pipeline that processes high-dimensional, noisy, and non-stationary biosignals. The fundamental challenge is cross-subject generalization: a model trained on 30 subjects and evaluated on held-out subjects faces substantial inter-individual variability in EEG patterns. Students must implement subject-independent train/test splits and discuss what the model has truly learned versus memorized.

Dataset: **DEAP Dataset** - 32-channel EEG recordings from 32 participants watching 40 one-minute music video clips, with self-reported valence and arousal labels.

04 Neural Machine Translation for a Low-Resource Language Pair

NLP

Supervised

Sequence-to-Sequence

Most neural machine translation (NMT) research focuses on high-resource language pairs (English–French, English–German) with hundreds of millions of parallel sentences. This project pushes students into the far more realistic and challenging low-resource regime, building a transformer-based encoder-decoder translation system for Urdu–English or Sindhi–English. Students will implement or fine-tune a multilingual pretrained model confront the real challenges of morphologically rich languages: tokenization strategy, out-of-vocabulary handling, and the limitations of BLEU score as an evaluation metric. A key deliverable is a qualitative error analysis that identifies systematic translation failures and discusses how data augmentation or transfer from related languages might address them.

Dataset: **OPUS Corpus** - A large collection of open multilingual parallel corpora including Urdu–English pairs from sources such as the Quran, GNOME, and CCAligned. Accessible at <https://opus.nlpl.eu>. Supplementary Urdu–English data also available via the UrduHack dataset on GitHub.

05 Anomaly Detection in Industrial Sensor Streams

Time Series

Unsupervised

Anomaly Detection

In real-world industrial settings, normal operating data is abundant but labeled anomaly examples are rare or nonexistent. This project challenges students to train a temporal autoencoder - either an LSTM-based or temporal convolutional autoencoder - exclusively on normal system behavior, then use reconstruction error as a proxy for anomaly likelihood at inference time. Students must design the architecture, train without any anomaly labels, and then carefully tune the decision threshold on a validation set. Key challenges include: choosing the right temporal window size, visualizing latent space structure using dimensionality reduction, and analyzing false positive rates at different threshold settings. The project produces concrete, measurable outcomes and forces students to think beyond supervised loss minimization.

Dataset: **SWAT Dataset** (Secure Water Treatment) - Sensor readings from a real water treatment testbed with labeled attack periods. Available from iTrust, Singapore: https://itrust.sutd.edu.sg/itrust-labs_datasets/. Alternative: **NASA SMAP/MSL** telemetry anomaly dataset at <https://github.com/khundman/telemanom>

06 Hate Speech Detection with Cross-Lingual Transfer

NLP

Supervised

Transfer Learning

Content moderation systems for hate speech are well-developed in English but severely lacking for most other languages, including Urdu and Arabic. This project tasks students with fine-tuning a multilingual transformer on English-language hate speech data, then evaluating zero-shot or few-shot transfer to a second language. This exposes students to a genuinely hard research problem: the model must transfer not just vocabulary knowledge, but cultural and contextual understanding of what constitutes hate speech across different linguistic communities. Evaluation must go beyond accuracy to include per-class F1, false positive rate for minority groups, and a fairness analysis. Students must also engage with dataset annotation disagreement and discuss how labeling biases are baked into the training data.

Dataset: **HatEval SemEval-2019** - Multilingual hate speech dataset in English and Spanish, available at <https://competitions.codalab.org/competitions/19935>.

07 Wildfire Risk Forecasting from Multi-Spectral Satellite Time Series

Remote Sensing

Supervised

Spatio-Temporal Prediction

Wildfires are accelerating in frequency and severity globally, and accurate short-range risk forecasting from satellite data has the potential to save lives and ecosystems. This project asks students to build a spatio-temporal deep learning model that ingests sequences of multi-spectral satellite image patches (including NDVI, land surface temperature, and soil moisture bands) and predicts next-week wildfire occurrence probability per grid cell. This is a genuinely hard modeling task: fires are spatially and temporally rare (extreme class imbalance), inputs are high-dimensional and multi-band, and spatial autocorrelation must be handled carefully during train/test splitting to prevent data leakage between neighboring grid cells.

Dataset: **NASA FIRMS** (Fire Information for Resource Management System) - Active fire detection data globally from MODIS and VIIRS sensors, freely downloadable at <https://firms.modaps.eosdis.nasa.gov>. Pre-processed wildfire prediction dataset also available on Kaggle: *Wildfire Prediction Dataset*.

08 Learned Image Compression with Convolutional Autoencoders

Computer Vision

Unsupervised

Generative Modeling

Classical image compression standards such as JPEG and WebP were designed by hand using signal processing principles. This project asks students to train a convolutional autoencoder to learn a compact, quantized latent representation of images end-to-end, purely from data. Students will implement an encoder-bottleneck-decoder pipeline, train with a combined rate-distortion loss (reconstruction quality vs. bit cost), and compare their model quantitatively against JPEG at different compression ratios using PSNR and SSIM metrics. Key technical challenges include implementing or approximating a differentiable quantization layer in the bottleneck, tuning the rate-distortion tradeoff hyperparameter, and visualizing what information the bottleneck retains versus discards. This project exposes students to information-theoretic thinking through an entirely unsupervised training objective.

Dataset: **CLIC** (Challenge on Learned Image Compression) - High-resolution images from professional and mobile photography, used in the annual compression challenge. Download at <https://www.compression.cc>. Alternative: **Kodak Lossless True Color Image Suite** - 24 high-quality reference images widely used in compression benchmarking, at <https://r0k.us/graphics/kodak/>

09 Multi-Region Tumor Segmentation in Brain MRI

Medical Imaging

Supervised

Semantic Segmentation

Accurately delineating tumor boundaries in brain MRI scans is a prerequisite for treatment planning and outcome prediction, yet it is a time-consuming and expert-dependent task. This project challenges students to train a U-Net or Attention U-Net to simultaneously segment three nested tumor sub-regions - the whole tumor, tumor core, and enhancing tumor - from four co-registered MRI modalities (T1, T1ce, T2, FLAIR). This is substantially harder than image classification: students must implement Dice loss (and explain why cross-entropy alone fails for highly imbalanced foreground/background segmentation), handle 3D or 2.5D volumetric input, and evaluate using the official BraTS evaluation protocol with per-region Dice and Hausdorff distance. Students are expected to compare at least two architectural variants and analyze qualitative failure cases.

Dataset: **BraTS 2020** (Brain Tumor Segmentation Challenge) - 369 multi-institutional MRI scans with expert-annotated tumor sub-region labels, in four modalities (T1, T1ce, T2, FLAIR). Fully open after registration at <https://www.med.upenn.edu/cbica/brats2020/data.html>

10 Graph Neural Networks for Drug–Drug Interaction Prediction

Graph Deep Learning

Supervised

Link Prediction

When two drugs are co-administered, their chemical interaction can produce adverse effects that neither drug causes alone. Predicting these drug–drug interactions (DDIs) is a critical problem in pharmacovigilance. This project introduces students to graph neural networks - a paradigm entirely distinct from CNNs and RNNs - by representing drug molecules as molecular graphs (atoms as nodes, bonds as edges) and training a GNN to predict adverse DDI types between drug pairs. Students will implement message-passing layers, graph-level pooling (mean, sum, or hierarchical), and a link prediction head on top of drug pair embeddings. The project demands learning new theory (graph convolution, message passing neural networks) rather than applying familiar architectures to a new dataset, making it one of the most pedagogically stretching options in this list.

Dataset: Processed graph-format DDI datasets also available on GitHub: *decagon-pytorch* and *SnapDDI*. Molecular graph generation: **RDKit** open-source cheminformatics library at <https://www.rdkit.org>

11 Diabetic Retinopathy Grading from Fundus Photographs

Medical Imaging

Supervised

Ordinal Classification

Diabetic retinopathy (DR) is the leading cause of preventable blindness worldwide, and its progression through five severity grades—from no DR to proliferative DR—is diagnosable from color fundus photographs. This project tasks students with training a CNN to perform ordinal five-class grading, where the key challenge is not just classification accuracy but understanding the inherent ordering of severity. Students must confront a severe class imbalance (most patients have mild or no disease), design appropriate loss functions such as weighted cross-entropy or ordinal regression losses, and apply targeted data augmentation for medical images (color jitter, random flips, zoom). A critical deliverable is a Grad-CAM analysis identifying which retinal structures—hemorrhages, microaneurysms, neovascularization—the model attends to at each severity level, and whether these align with clinical guidelines. Students must also report the quadratic weighted kappa, the official competition metric, rather than relying on accuracy alone.

Dataset: **APTOS 2019 Blindness Detection** (Kaggle) — 3,662 high-resolution fundus images labeled with DR severity grades 0–4, collected from rural India. Available at

<https://www.kaggle.com/competitions/aptos2019-blindness-detection>.

Supplementary: **EyePACS** dataset with 88,000 images at

<https://www.kaggle.com/c/diabetic-retinopathy-detection>.

12 Arrhythmia Classification from 12-Lead ECG Signals

Time Series

Supervised

Multi-label Classification

An electrocardiogram (ECG) records the electrical activity of the heart across multiple leads simultaneously and encodes diagnostic information about a wide range of cardiac conditions. This project asks students to build a deep learning model—a 1D ResNet or a temporal convolutional network—to classify 12-lead ECG recordings into multiple co-occurring arrhythmia types (multi-label, not multi-class). The core technical challenges are several: the input is a multi-channel 1D signal of fixed length, multiple labels can be simultaneously present for one patient, and the dataset exhibits long-tailed class distributions where rare but dangerous arrhythmias are severely underrepresented. Students must implement appropriate multi-label losses (binary cross-entropy with class weights), justify their temporal resolution and windowing choices, and report per-class AUROC across all rhythm types—the standard clinical evaluation metric. Students are expected to discuss the gap between model performance on held-out test data and what would be required for clinical deployment.

Dataset: PhysioNet/CinC Challenge 2021 — Over 88,000 twelve-lead ECG recordings from six international institutions, annotated with 26 cardiac rhythm classes. Available at

<https://physionet.org/content/challenge-2021/>.

Simpler alternative for initial exploration: **MIT-BIH Arrhythmia Database** at

<https://physionet.org/content/mitdb/>.

13 Facial Action Unit Detection with Multi-label CNNs

Computer Vision

Supervised

Multi-label Classification

The Facial Action Coding System (FACS) decomposes human facial expressions into anatomically defined muscle movements called Action Units (AUs), such as AU1 (inner brow raise) or AU12 (lip corner pull). Automatically detecting which AUs are active in a face image is the foundation for expression recognition, pain assessment, and deception detection. This project tasks students with building a CNN-based multi-label detector for a subset of the most common AUs. The key difficulty is that AUs are not mutually exclusive—multiple can fire simultaneously—and many are visually subtle, requiring the model to attend to localized facial regions. Students must implement attention mechanisms or region-of-interest pooling to focus on relevant facial landmarks, apply strong face-specific augmentation, and analyze common failure modes such as identity leakage (the model memorizing subject identities instead of AU patterns). The project requires careful subject-independent cross-validation to prevent identity leakage from inflating reported performance.

Dataset: BP4D-Spontaneous — 41 subjects performing 10 emotion-eliciting tasks, with frame-level AU annotations for 12 AUs, totaling over 140,000 labeled frames. Available from the Binghamton University Facial Expression research group upon request. Alternative: **DISFA** dataset with 27 subjects and per-frame AU intensity labels at <http://mohammadmahoor.com/disfa/>.

14 Video Action Recognition with 3D Convolutional Networks

Computer Vision

Supervised

Video Understanding

Recognizing human actions in video requires integrating spatial appearance information across time—a fundamentally different problem from image classification. This project asks students to implement and compare two paradigms: a 3D CNN (C3D or R(2+1)D) that jointly models spatial and temporal features across a short video clip, and a two-stream network that processes appearance (RGB frames) and motion (optical flow) in separate branches. Students must handle the unique en-

engineering challenges of video data: memory-efficient clip sampling strategies, temporal data augmentation (speed perturbation, temporal jittering), and the computational cost of 3D convolutions. The ablation study is central to this project: students must isolate the contribution of temporal modeling by comparing 3D CNNs against a 2D CNN baseline that simply averages per-frame predictions, and quantify how much the motion stream adds over appearance alone. Students should also analyze which action categories benefit most from temporal context versus those solvable from a single frame.

Dataset: **UCF-101** — 13,320 video clips from 101 human action categories in realistic conditions. Fully open at <https://www.crcv.ucf.edu/data/UCF101.php>. Alternative with more diversity: **HMDB-51** at <https://serre-lab.clps.brown.edu/resource/hmdb-a-large-human-motion-database/>. For teams with GPU resources, **Kinetics-400** (a 400-class subset) is available on Academic Torrents.

15 Knowledge Distillation for Efficient On-Device Image Classification

Computer Vision

Model Compression

Supervised

Deploying deep learning models on resource-constrained devices — mobile phones, embedded sensors, edge nodes — requires models that are accurate, fast, and small. Knowledge distillation offers an elegant solution: train a compact *student* network to mimic the output distribution (soft labels) of a large pretrained *teacher* network, recovering much of the teacher’s accuracy at a fraction of the computational cost. This project asks students to implement the Hinton et al. distillation objective (a combination of hard-label cross-entropy and KL divergence against softened teacher logits), vary the temperature hyperparameter, and measure the student-teacher accuracy gap as a function of the student’s depth and width. Crucially, students must conduct a systematic study of *what* the teacher is transferring: by comparing student models trained on hard labels alone versus soft labels, they must explain why soft probability distributions over incorrect classes (the “dark knowledge”) improve generalization. Advanced extensions include intermediate feature-level distillation and comparing distillation against pruning and quantization as complementary compression strategies. This project builds deep intuition for the bias-variance-efficiency tradeoffs that are central to the course.

Dataset: **CIFAR-100** — 60,000 images across 100 fine-grained classes (500 training, 100 test per class), bundled in PyTorch/TensorFlow. Teacher model: pretrained **ResNet-110** or **WideResNet-40**; student models: **ResNet-20** or **MobileNetV2**. For a more challenging setting, **TinyImageNet** (200 classes, 100,000 images at 64×64) is available at:

<http://cs231n.stanford.edu/tiny-imagenet-200.zip>.

Supplementary Reading List

The following reading lists are provided for projects that draw on techniques **not covered in the AI-600 course outline**. Students undertaking these projects are expected to engage with the relevant papers as part of their literature review. Readings are ordered from foundational to advanced within each project.

07 Wildfire Risk Forecasting — Spatio-Temporal Modeling

This project requires modeling spatial correlations across image grids over time, which goes beyond the sequence-to-sequence and encoder–decoder content covered in the course. The following papers introduce the convolutional LSTM architecture and its application to spatio-temporal prediction problems.

- Shi, X. et al. (2015). **Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting**. *NeurIPS 2015*. [arXiv:1506.04214](#) — Introduces ConvLSTM, the core architecture for grid-structured temporal prediction. A direct methodological foundation for this project.
- Wang, Y. et al. (2023). **PreDiff: Precipitation Nowcasting with Latent Diffusion Models**. *NeurIPS 2023*. [arXiv:2307.10422](#) — Shows how diffusion models extend to spatio-temporal prediction; relevant for teams considering generative baselines.

09 Brain MRI Segmentation — U-Net and Volumetric Architectures

The U-Net architecture and volumetric (3D) processing extend beyond the CNN backbone material (LeNet, VGG, ResNet) covered in the course. The papers below establish the foundational segmentation architectures and the Dice loss function that this project requires.

- Ronneberger, O., Fischer, P., & Brox, T. (2015). **U-Net: Convolutional Networks for Biomedical Image Segmentation**. *MICCAI 2015*. [arXiv:1505.04597](#) — The original U-Net paper; required reading. Explains skip connections and the encoder–decoder design for dense prediction.
- Milletari, F., Navab, N., & Ahmadi, S.-A. (2016). **V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation**. *3DV 2016*. [arXiv:1606.04797](#) — Extends U-Net to 3D and introduces the Dice loss function. Essential for understanding why cross-entropy fails on imbalanced foreground/background segmentation.
- Oktay, O. et al. (2018). **Attention U-Net: Learning Where to Look for the Pancreas**. *MIDL 2018*. [arXiv:1804.03999](#) — Introduces the attention gate mechanism that softly suppresses irrelevant background activations; directly relevant to the Attention U-Net variant required by this project.

10 Drug–Drug Interaction — Graph Neural Networks

Graph neural networks represent a paradigm entirely absent from the AI-600 syllabus. This project requires students to learn graph convolution and message passing from scratch. The papers below form a compact but complete introduction to the field.

- Kipf, T. N., & Welling, M. (2017). **Semi-Supervised Classification with Graph Convolutional Networks**. *ICLR 2017*. [arXiv:1609.02907](#) — The foundational GCN paper. Derives graph convolution from spectral graph theory and demonstrates node classification on citation networks. Start here.
- Gilmer, J. et al. (2017). **Neural Message Passing for Quantum Chemistry**. *ICML 2017*. [arXiv:1704.01212](#) — Introduces the message-passing neural network (MPNN) framework that unifies most GNN variants under a single formalism. Required for understanding the architecture design space.
- Xu, K. et al. (2019). **How Powerful are Graph Neural Networks?** *ICLR 2019*. [arXiv:1810.00826](#) — Theoretical analysis of GNN expressiveness via the Weisfeiler–Lehman isomorphism test. Introduces the Graph Isomorphism Network (GIN). Important for ablation design.

13 Facial Action Unit Detection — FACS and Attention Localization

This project requires knowledge of the Facial Action Coding System and of region-specific attention mechanisms that focus convolutional features on anatomically meaningful facial areas. Neither is covered in the course.

- Shao, Z. et al. (2021). **JAA-Net: Joint Facial Action Unit Detection and Face Alignment via Adaptive Attention**. *International Journal of Computer Vision*, 129(2). [arXiv:2003.08834](#) — Proposes joint alignment and AU detection with region-level attention maps; one of the strongest baselines for this task and a good model for architecture design.
- Li, W. et al. (2019). **Self-Supervised Representation Learning from Videos for Facial Action Unit Detection**. *CVPR 2019*. [CVF Open Access](#) — Demonstrates how temporal self-supervision improves AU features; relevant for teams considering extensions beyond single-frame prediction.
- Zhao, K. et al. (2016). **Deep Region and Multi-Label Learning for Facial Action Unit Detection**. *CVPR 2016*. [CVF Open Access](#) — Introduces region-of-interest cropping aligned with facial landmarks as a preprocessing step; useful for understanding the spatial localization challenge.

14 Video Action Recognition — 3D CNNs and Two-Stream Networks

The course covers 2D spatial CNNs and 1D temporal sequence models, but does not address the fusion of spatial and temporal streams in video, nor 3D convolutions and optical flow estimation. The following papers establish the two main paradigms students will implement and compare.

- Tran, D. et al. (2015). **Learning Spatiotemporal Features with 3D Convolutional Networks**. *ICCV 2015*. [arXiv:1412.0767](#) — The C3D paper; extends 2D convolution to 3D for joint spatial–temporal feature learning on short video clips. Essential for the 3D CNN branch of this project.
- Tran, D. et al. (2018). **A Closer Look at Spatiotemporal Convolutions for Action Recognition**. *CVPR 2018*. [arXiv:1711.11248](#) — Introduces the R(2+1)D factorization, separating 3D convolution into a 2D spatial and 1D temporal conv. Important for understanding the design tradeoffs students must ablate.
- Carreira, J., & Zisserman, A. (2017). **Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset**. *CVPR 2017*. [arXiv:1705.07750](#) — The I3D paper; inflates 2D ImageNet-pretrained filters into 3D. Provides important context on pretraining strategies for video models.

15 Knowledge Distillation — Model Compression Theory

Knowledge distillation is not covered in the AI-600 syllabus, which addresses regularization and capacity control but not post-hoc compression. The papers below span the original formulation, intermediate-layer variants, and a survey connecting distillation to pruning and quantization.

- Hinton, G., Vinyals, O., & Dean, J. (2015). **Distilling the Knowledge in a Neural Network**. *NeurIPS 2014 Deep Learning Workshop*. [arXiv:1503.02531](#) — The foundational paper. Introduces soft targets, temperature scaling, and the concept of “dark knowledge.” Start here before any implementation work.
- Romero, A. et al. (2015). **FitNets: Hints for Thin Deep Nets**. *ICLR 2015*. [arXiv:1412.6550](#) — Extends distillation beyond output logits to intermediate feature maps (“hints”), enabling compression into thinner but deeper student networks. Relevant for the feature-level distillation extension.
- Cho, J. H., & Hariharan, B. (2019). **On the Efficacy of Knowledge Distillation**. *ICCV 2019*. [arXiv:1910.01348](#) — Provides a rigorous empirical analysis of when and why distillation helps or fails; important context for designing a credible ablation study.
- Gou, J. et al. (2021). **Knowledge Distillation: A Survey**. *International Journal of Computer Vision*, 129(6). [arXiv:2006.05525](#) — Comprehensive survey covering response-based, feature-based, and relation-based distillation, with connections to pruning and quantization. Useful for literature review framing.